

Asymmetric On-Policy Distillation: Bridging Exploitation and Imitation at the Token Level

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Abstract

On-policy distillation (OPD) trains a student on its own trajectories with token-level teacher feedback and often outperforms off-policy distillation and standard reinforcement learning. However, we find that its standard advantage-weighted policy gradient suffers from three structural weaknesses, including high variance updates, vanishing gradients in zero-advantage regions, and exploration bottlenecks when corrective signals are insufficient. We therefore propose **Asymmetric On-Policy Distillation (AOPD)**, which replaces ineffective negative reinforcement with localized divergence minimization in non-positive advantage regions while preserving positive reinforcement learning. Experiments on mathematical reasoning benchmarks show that AOPD consistently outperforms standard OPD, with average gains of 4.09 / 8.34 under strong / weak initialization, respectively. AOPD also maintains higher policy entropy during training and better capability retention during sequential tool-use adaptation.

1 Introduction

The strong capabilities of large language models (LLMs) (OpenAI, 2025; DeepSeek-AI et al., 2025; Yang et al., 2025) come at steep computational cost. Knowledge distillation (Hinton et al., 2014) offers a principled path to compress these capabilities into smaller models by training a student to mimic a teacher’s output distribution (Zhu et al., 2024). Classical off-policy distillation has achieved considerable success (Xu et al., 2024; Hsieh et al., 2023; Tian et al., 2025). However, this paradigm carries a well-known vulnerability that the student encounters exposure bias (Bengio et al., 2015) at inference time after training exclusively on the teacher’s trajectories.

On-policy distillation, as advanced by the work of Thinking Machines Lab (Lu and Lab, 2025), resolves this mismatch by leveraging teacher probabilities on student-generated trajectories to con-

struct dense, token-level reward signals under a policy gradient objective. However, we observe that this reward formulation exhibits several limitations. The calculated updates exhibit substantial variance in negative advantage regions. Simultaneously, a vast number of generated positions yield an advantage of zero, causing the policy gradients to vanish almost entirely. Beyond these issues, the learning paradigm relies heavily on autonomous exploration, rendering the model blind to correct alternatives outside its prior.

To address these limitations, we bridge the principles of reinforcement learning and supervised learning to propose Asymmetric On-Policy Distillation (AOPD). Figure 1 provides an illustration of the two learning modes in AOPD, namely exploitation and imitation. Specifically, when a generated token receives a negative score, rather than forcing the model to learn through self-exploration, AOPD directly learns the teacher model’s target distribution conditioned on the current on-policy trajectory prefix. This asymmetric design allows the student to retain the benefit of exploitation in reinforcement learning while receiving stronger correction at genuine optimization bottlenecks.

We evaluate AOPD on competition-level mathematical reasoning benchmarks. Across model scales and warm-up settings, AOPD delivers stronger and more robust performance than existing baselines. The localized guidance also leads to better capability retention in continual learning. Our main contributions are summarized as follows:

- We analyze on-policy distillation in negative and zero advantage regions and show that the standard advantage-based update provides limited learning signal for effective correction.
- We propose Asymmetric On-Policy Distillation (AOPD), a token-level training framework that adaptively shifts from advantage-weighted policy gradient to localized distri-

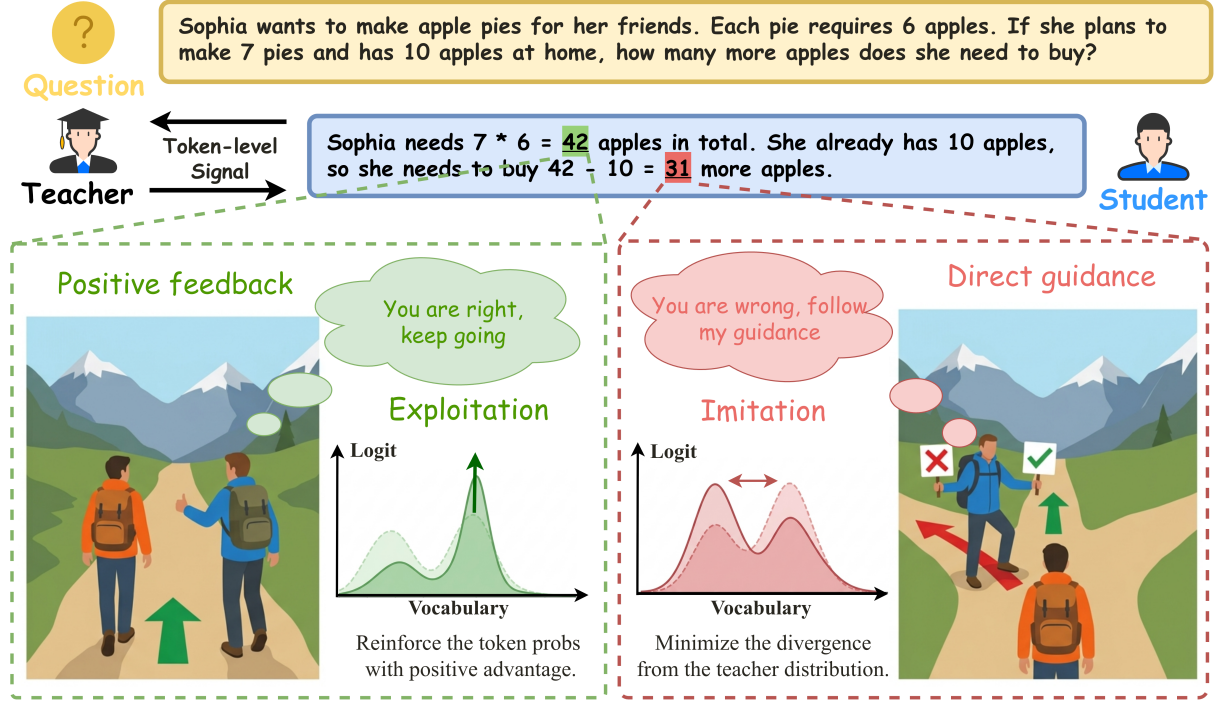


Figure 1: **Overview of Asymmetric On-Policy Distillation (AOPD).** The asymmetry comes from using two different learning modes on student-generated trajectories: preserving exploitation on aligned positions and invoking teacher guidance on bottleneck positions. **Left (Exploitation):** When the student’s reasoning aligns with the teacher, AOPD reinforces successful exploration. **Right (Imitation):** When the student encounters an exploration black hole, AOPD actively switches to directed teacher guidance by minimizing the distributional divergence.

bution matching with the teacher when the advantage signal carries limited information.

- Experiments on mathematical reasoning benchmarks across model scales and warm-up settings show that AOPD consistently improves over baselines, especially under weak initialization, and better preserves prior capabilities during sequential tool-use adaptation.

2 Related Work

The foundation of knowledge distillation was established by Hinton et al. (2014), who showed that a compact student can learn from a larger teacher by matching its softened output logits. For sequence generation, Kim and Rush (2016) extended this idea to Sequence-Level Knowledge Distillation (SeqKD), aligning the student with the teacher’s sequence-level distribution. However, all these classical methods are trained off-policy on teacher-generated trajectories and therefore remain vulnerable to exposure bias (Bengio et al., 2015).

To mitigate this mismatch, recent work has shifted toward on-policy distillation, where the student learns on its own rollouts with teacher

feedback. GKD (Agarwal et al., 2024) formalizes this as minimizing generalized f -divergences on student-generated trajectories. Gu et al. (2024) and Lu and Lab (2025) formulate on-policy distillation as a reinforcement learning objective with dense token-level rewards derived from the teacher. Beyond teacher alignment, ExOPD (Yang et al., 2026) extrapolates reward scaling to surpass the teacher’s capabilities. Subsequent work removes the reliance on external teachers by distilling from the model’s own distributions. OPSD (Zhao et al., 2026) and its long-context extension OPSDL (Zhang et al., 2026b) leverage ground-truth feedback on self-generated traces, while SDFT (Shenfeld et al., 2026) mitigates catastrophic forgetting under the same paradigm.

3 Preliminaries

We formalize the training procedure of on-policy distillation below. Given an input prompt x , let y denote a complete response sampled from the student policy. OPD optimizes the reverse Kullback-Leibler (KL) divergence with reinforcement learning based on the K1 estimator. At step t , let $c_t \triangleq (x, y_{<t})$ denote the current context, and let

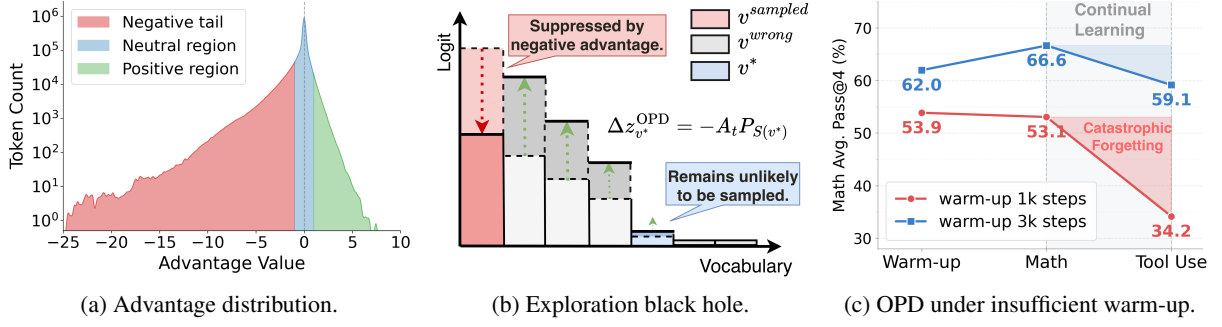


Figure 2: Observation and analysis of On-Policy Distillation.

$P_T(\cdot | c_t)$ and $P_S(\cdot | c_t)$ denote the teacher and student conditional distributions, respectively. The advantage of token $y_t \sim P_S(\cdot | c_t)$ is defined as

$$A_t = \text{sg}[\log P_T(y_t | c_t) - \log P_S(y_t | c_t)], \quad (1)$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operator. The corresponding training loss is

$$\mathcal{L}_{\text{OPD}} = -\mathbb{E} \left[\frac{1}{|y|} \sum_{t=1}^{|y|} A_t \log P_S(y_t | c_t) \right]. \quad (2)$$

Motivated by [Zhu et al. \(2026\)](#) and [Tang et al. \(2025\)](#), which highlight the distinct roles of positive and negative samples in reinforcement learning, we decompose OPD into positive and negative reinforcement at the token level. Let S^+ and S^- denote the sets of token positions with positive and negative advantages, respectively, and let y_t^+ and y_t^- denote the corresponding sampled tokens. The OPD loss can be written as

$$\mathcal{L}_{\text{OPD}} = \underbrace{-\mathbb{E} \left[\frac{1}{|y|} \sum_{t \in S^+} A_t \log P_S(y_t^+ | c_t) \right]}_{\mathcal{L}_{\text{Pos}}} - \underbrace{\mathbb{E} \left[\frac{1}{|y|} \sum_{t \in S^-} A_t \log P_S(y_t^- | c_t) \right]}_{\mathcal{L}_{\text{Neg}}}. \quad (3)$$

Here, $\mathcal{L}_{\text{Pos}}$ corresponds to positive reinforcement, which reinforces teacher-favored sampled tokens and corresponds to exploitation. In contrast, $\mathcal{L}_{\text{Neg}}$ corresponds to negative reinforcement, which suppresses teacher-disfavored sampled tokens and promotes exploration.

4 Limitations of Negative Reinforcement in OPD

Despite its intuitive formulation, standard OPD remains subject to several challenges in practice.

Through theoretical analysis and experimental validation, we identify three characteristic limitations of OPD in negative reinforcement.

Heavy Tails in Negative Advantages. Figure 2a shows that the advantage distribution is highly asymmetric. Most tokens cluster near zero while the negative side exhibits a substantially broader tail. This extreme variance originates directly from the logarithmic nature of the advantage formulation. The difference between the log probability of the teacher and the student amplifies exponentially as the student probability approaches zero. Because the update magnitude is directly scaled by this unbounded scalar, the mechanism artificially inflates gradient variance and renders the overall optimization fragile.

Stagnation at Zero Advantages. A second limitation arises in the neutral regime. As illustrated by the distribution peak in Figure 2a, the vast majority of sampled tokens cluster around an advantage of zero. The standard OPD objective consequently yields an almost vanishing update. Even when the advantage of the sampled token is zero, the teacher still provides a rich conditional distribution over the vocabulary, offering fine-grained preferences over alternative reasoning paths. However, OPD compresses this supervision into a scalar signal that is close to zero.

Exploration Black Hole. A fundamental optimization problem arises in negative reinforcement, where the update of OPD is restricted in both direction and magnitude. To understand this behavior, we examine the OPD update at the logit level. Let z_v denote the student logit for token v under context c_t , so that $P_S(v | c_t) = \text{softmax}(z)_v$. By the softmax Jacobian, the update for any token v is given by:

$$\Delta z_v^{\text{OPD}} \propto A_t (\mathbb{I}(v = y_t) - P_S(v | c_t)). \quad (4)$$

When $A_t < 0$, the sampled token is penalized, and for every unsampled token $v \neq y_t$, the update becomes

$$\Delta z_v^{\text{OPD}} \propto -A_t P_S(v | c_t) > 0. \quad (5)$$

Hence, the released probability mass is redistributed strictly according to the current prior of the student. As illustrated in Figure 2b, for an essential token v^* with negligible prior probability $P_S(v^* | c_t) \approx 0$, the corrective update is correspondingly negligible even if that token is favored by the teacher. OPD therefore suppresses the sampled mistake without effectively promoting the correct alternative, trapping the model in an exploration black hole. This pathology is more severe for student models with weaker foundational capabilities, as essential tokens are more likely to reside in the low-probability tail of the student prior. Figure 2c illustrates this effect by comparing OPD models with different steps of SFT warm-up on OpenThoughts (Guha et al., 2026), followed by training on mathematical reasoning tasks, and then continual training on tool-use tasks. Models with fewer warm-up steps begin from a weaker starting point, making exploration black holes more likely and thereby undermining both reasoning improvement and subsequent continual learning.

These observations suggest that tokens in non-positive regions should not be optimized in the same way as positive reinforcement, motivating an asymmetric training rule.

5 Asymmetric On-Policy Distillation

5.1 The AOPD Framework

To systematically resolve the structural vulnerability of standard on-policy distillation, we propose Asymmetric On-Policy Distillation (AOPD). This framework dynamically modulates the learning paradigm at the token level, shifting between reinforcement learning for exploitation and soft-label supervised learning for imitation based on the student’s immediate capability.

We compute the token-level advantage A_t using Eq. 1 like standard OPD. As discussed in Section 4, the main optimization difficulties of OPD are concentrated in negative and zero-advantage regions. We therefore replace the update in non-positive regions with forward KL guidance computed on the teacher-defined support. For each position t , we define the teacher support as

$$S_t = \text{TopK}(P_T(\cdot | c_t), K), \quad (6)$$

where S_t contains the K highest-probability tokens under the teacher distribution. On this support, the forward KL guidance loss is instantiated as

$$\mathcal{L}_t^{\text{FKL}} = \frac{1}{K} \sum_{v \in S_t} P_T(v | c_t) \left(\log P_T(v | c_t) - \log P_S(v | c_t) \right). \quad (7)$$

Accordingly, the AOPD objective is defined as

$$\mathcal{L}_{\text{AOPD}} = \mathbb{E} \left[\frac{1}{|y|} \sum_{t \in S^{\leq 0}} \mathcal{L}_t^{\text{FKL}} \right] + \mathcal{L}_{\text{Pos}}, \quad (8)$$

where $S^{\leq 0}$ denotes the zero and negative advantage token set.

AOPD uses zero advantage as the default intervention boundary. To define the intervention location in a more general form, we further introduce a threshold parameter τ . Considering the logarithmic nature of the standard advantage formulation and its unbounded variance, we use the bounded probability difference to determine whether to trigger teacher intervention. Specifically, we define the token-level mask as:

$$G_t = \mathbb{I}(P_T(y_t | c_t) - P_S(y_t | c_t) \leq \tau), \quad (9)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. The generalized AOPD objective is then defined as:

$$\mathcal{L}_{\text{AOPD}} = \mathbb{E} \left[\frac{1}{|y|} \sum_{t=1}^{|y|} \left(G_t \mathcal{L}_t^{\text{FKL}} + (1 - G_t) \mathcal{L}_t^{\text{OPD}} \right) \right]. \quad (10)$$

Setting $\tau = -1$ disables intervention for all tokens and reduces AOPD to standard OPD, while setting $\tau = 1$ applies supervised distribution matching everywhere and recovers a GKD objective. In our method, we set $\tau = 0$, so that intervention is applied exactly to the non-positive regime. The effects of alternative choice of τ and other intervention locations are further analyzed in Section 5.3 and Section 6.4.

5.2 Truncated Forward-KL Guidance

In principle, the intervention objective should minimize the divergence between the teacher and the student over the full vocabulary. The divergence

family can be written in the general form of Jensen–Shannon(JS) divergence as

$$\begin{aligned} \text{JSD}_\beta(P_T \parallel P_S) &= \beta D_{\text{KL}}(P_T \parallel P_M) \\ &\quad + (1 - \beta) D_{\text{KL}}(P_S \parallel P_M), \end{aligned} \quad (11)$$

where $P_M = \beta P_T + (1 - \beta) P_S$ and $\beta \in [0, 1]$. Here, $\beta = 1$ corresponds to forward KL, while $\beta = 0$ corresponds to reverse KL.

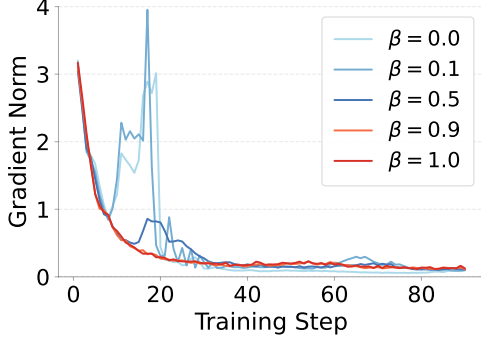


Figure 3: Gradient norm under different values of β .

However, evaluating such a divergence at every intervened position is computationally prohibitive in large-vocabulary LLM training. We therefore instantiate the guidance term on the teacher-selected top- K support in Eq. 6. Under top- K truncation, the objective is no longer a full-vocabulary divergence, but a correction objective defined on a teacher-selected domain. In this setting, the teacher specifies both the candidate tokens in the optimization domain and the reference distribution within that domain. Therefore, the intervention objective should remain teacher-centered after truncation. Forward KL is the natural choice because it preserves the same teacher-conditioned measure in both support construction and loss weighting, whereas reverse KL would reweight the update according to the student’s current distribution on a domain selected by the teacher.

Empirical evidence further favors placing more emphasis on the teacher distribution. Figure 3 shows that as the divergence becomes more biased toward the teacher distribution, the optimization exhibits more stable gradient behavior. In particular, $\beta = 0.9$ and 1 lead to the smoothest training dynamics. This trend is consistent with the performance comparison in Section 6.4. Combined with the support-consistency argument above, these results motivate our choice of $\beta = 1$ in AOPD. Further analyses of gradient behavior and training metrics are provided in Appendix B.

5.3 Why AOPD Resolves the Three Bottlenecks of OPD

We now explain why the single confidence-based rule in AOPD is sufficient to unify the solutions for the three optimization bottlenecks identified in Section 4. By setting $\tau = 0$, the core mechanism is to switch from policy gradient to direct distribution matching exactly where $A_t \leq 0$.

(1) Eliminating high-variance negative advantages. As analyzed in Section 4, standard OPD suffers from heavy-tailed gradient noise when $A_t < 0$ because the scalar advantage is unbounded. AOPD eliminates this instability by halting the policy gradient update in these regions and replacing it with divergence guidance. For the forward KL case, the logit-level correction is instead governed by the bounded distribution gap:

$$\Delta z_v^{\text{AOPD}} \propto P_T(v \mid c_t) - P_S(v \mid c_t). \quad (12)$$

Constrained by the probability simplex, this signal naturally caps the maximum update magnitude. Figure 4a confirms this effect, showing that AOPD maintains a substantially smaller gradient norm than OPD throughout training.

(2) Recovering gradients in zero-advantage regions. Recall from Section 4 that informative signal is lost when $A_t \approx 0$ because the advantage-weighted update vanishes. AOPD avoids this stagnation because its intervention relies on the discrepancy between the conditional distributions, rather than the scalar A_t . Consequently, even in neutral regions where the sampled token provides only a weak learning signal, the student can still receive meaningful corrections on the teacher-defined support. Figure 4b directly supports this by evaluating AOPD-ZERO, a variant that applies KL-divergence intervention only to tokens whose advantages are zero. Compared with standard OPD, this targeted replacement already yields clear gains, confirming that neutral regions contain useful teacher signals that are discarded by the policy gradient method. Furthermore, as training progresses and the student’s output aligns more closely with the teacher, an increasing number of generated tokens begin to cluster around $A_t \approx 0$. This trend is explicitly captured in Figure 4c where the proportion of tokens triggering AOPD-ZERO intervention steadily rises from near zero to roughly 30%, ensuring that learning does not prematurely stagnate.

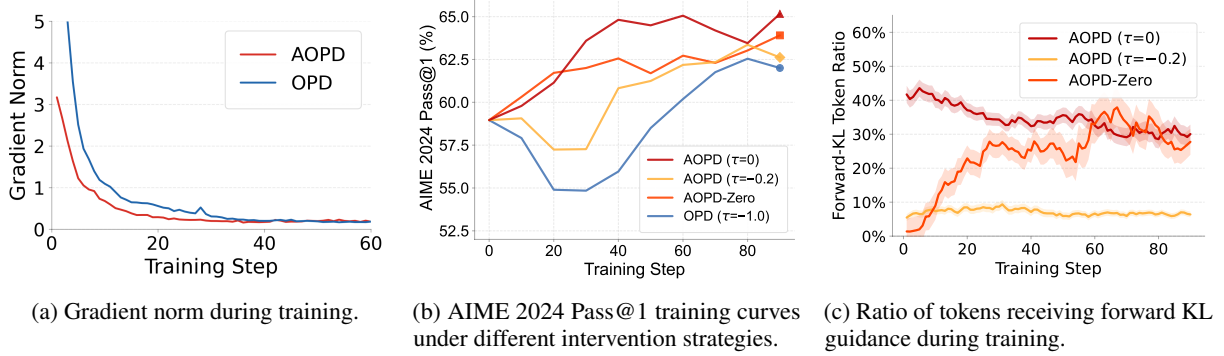


Figure 4: Training dynamics under different divergence-guidance strategies.

(3) Escaping exploration bottlenecks through teacher guidance. When trapped in an exploration black hole, OPD merely suppresses the sampled mistake without promoting the correct, unsampled reasoning path, forcing the model to blindly explore. AOPD resolves this by explicitly injecting the missing directional signal. Any token favored by the teacher but underestimated by the student immediately receives a positive correction proportional to their probability gap. By directly matching the teacher’s target distribution on the on-policy prefix, AOPD repairs optimization bottlenecks instantly rather than waiting for the student to discover the correct token. This is further evidenced in Figure 4b by the $\tau = -0.2$ variant, which applies forward KL guidance only when the teacher–student gap is sufficiently negative. As shown in Figure 4c, this variant intervenes on fewer than 10% of all generated tokens. However, even this sparse intervention over strongly negative regions already yields clear gains over standard OPD, confirming that targeted teacher distribution guidance is highly efficient at rescuing optimization from exploration black holes.

5.4 Localized Supervision Preserves the Policy Space

Beyond resolving the three optimization bottlenecks of OPD, the localized supervision in AOPD also leads to a training dynamic that avoids excessive reshaping of the full policy. Figure 4c shows that the fraction of tokens receiving forward KL guidance is limited and gradually decreases during training from roughly 40% to 30%, which indicates that direct distributional correction remains confined to a shrinking subset of positions. Figure 5 shows that AOPD maintains a substantially higher policy entropy throughout training than OPD and

remains above GKD over the course of optimization, which indicates that its policy stays less concentrated during training.

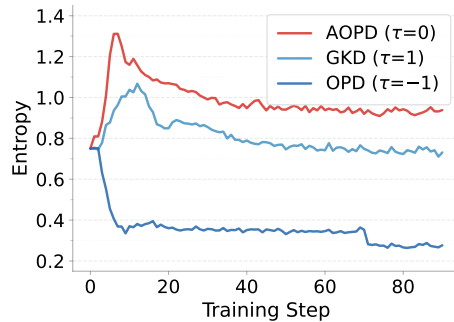


Figure 5: Policy entropy during training.

These two observations jointly suggest that AOPD does not continuously reshape the full policy during training. Instead, it preserves a broader policy space while restricting teacher intervention to a limited set of difficult positions. This training behavior is consistent with the continual learning results in Section 6.3, where AOPD retains prior reasoning ability better after tool-use adaptation.

6 Experiments

6.1 Experimental Setup

Models. We evaluate our approach under two distinct distillation settings: (1) distilling from Qwen3-32B to Qwen3-8B-Base, and (2) distilling from Qwen3-8B to Qwen3-4B-Base (Yang et al., 2025). These settings provide challenging scenarios to validate our method across different parameter scales.

Training Data. For the warm-up stage, we utilize the OpenThoughts dataset (Guha et al., 2026) for general and mathematical reasoning. During the on-policy distillation phase, mathematical reasoning is conducted on DeepMath (He et al., 2025). For

Table 1: Main results across multiple math reasoning benchmarks. Baseline capabilities before on-policy distillation are given by the SFT Checkpoint. We compare AOPD against SeqKD, OPD, GKD and ExOPD. **Bold** indicates the best result within each configuration block, while underlined values denote the second best.

Method	AIME 2024		AIME 2025		HMMT 2025(Feb)		Average	
	Pass@1	Pass@4	Pass@1	Pass@4	Pass@1	Pass@4	Pass@1	Pass@4
Qwen3-4B-Base								
– 3K SFT Warm-up								
SFT Checkpoint	52.29	71.90	46.25	64.82	27.08	38.97	41.87	58.56
+ SeqKD	42.50	65.23	40.00	60.09	24.58	38.01	35.69	54.44
+ OPD	57.08	76.69	51.88	68.12	32.92	47.65	47.29	64.15
+ ExOPD	<u>59.17</u>	73.99	51.25	66.59	<u>32.08</u>	47.30	<u>47.50</u>	62.63
+ GKD	58.13	<u>76.61</u>	50.42	<u>69.12</u>	29.79	45.39	46.11	63.71
+ AOPD	61.04	75.08	54.37	69.73	<u>32.08</u>	<u>47.40</u>	49.16	<u>64.07</u>
Qwen3-8B-Base								
– 1K SFT Warm-up								
SFT Checkpoint	45.21	68.29	39.79	58.62	24.58	34.68	36.53	53.86
+ SeqKD	40.21	68.08	30.63	50.00	20.62	36.78	26.11	42.87
+ OPD	43.13	67.38	38.12	56.84	23.13	34.95	34.79	53.06
+ ExOPD	45.83	72.31	36.04	53.86	24.37	38.02	35.41	54.73
+ GKD	54.79	77.92	<u>42.08</u>	<u>63.69</u>	<u>25.83</u>	<u>40.21</u>	<u>40.90</u>	<u>60.61</u>
+ AOPD	<u>53.75</u>	<u>77.39</u>	45.21	66.86	30.42	46.73	43.13	63.66
– 3K SFT Warm-up								
SFT Checkpoint	58.96	74.95	47.71	66.73	31.04	44.19	45.90	61.96
+ SeqKD	51.04	76.56	40.00	56.98	25.42	43.80	38.82	59.11
+ OPD	61.46	78.61	<u>52.29</u>	<u>70.44</u>	<u>34.17</u>	<u>50.86</u>	49.31	66.64
+ ExOPD	61.67	77.55	48.75	65.83	<u>34.17</u>	50.22	48.20	64.53
+ GKD	<u>66.25</u>	80.65	48.54	70.56	<u>34.17</u>	49.67	<u>49.65</u>	<u>66.96</u>
+ AOPD	66.87	<u>79.58</u>	55.00	70.28	38.33	52.42	53.40	67.43

the tool-use task in the continual learning stage, we use ToolAlpaca (Tang et al., 2023).

Benchmarks. We evaluate mathematical reasoning on AIME 2024, AIME 2025, and HMMT 2025(Feb). For tool-use evaluation, we use the ToolAlpaca test set. We report Pass@1/4 for reasoning benchmarks, and tool-call success rate for tool-use evaluation.

Training Pipeline. We first perform warm-up training with 128 batch size on the base models. Specifically, the Qwen3-4B-Base model is warmed up for 3,000 steps. For the Qwen3-8B-Base model, we warm it up for 3,000 steps, alongside an experimental setting of 1,000 steps to yield variants with distinct foundational capabilities for robustness assessment. During the subsequent distillation phase, we uniformly set the learning rate to $1e-5$ and batch size to 512. We set the teacher support size K to 32 in AOPD and GKD. Models are trained for 90 steps on mathematical tasks, followed by 40 steps on tool-call tasks for continual learning.

Baselines. We compare AOPD against four baselines: (i) **SeqKD** (Kim and Rush, 2016), which continues supervised fine-tuning on the DeepMath dataset, utilizing sequence-level knowledge distil-

lation with trajectories generated by the teacher model; (ii) **OPD** (Lu and Lab, 2025), standard on-policy distillation using K1-estimated KL divergence as the per-token advantage for policy gradient optimization; (iii) **GKD** (Agarwal et al., 2024), which minimizes Jensen-Shannon divergence on student-generated trajectories; and (iv) **ExOPD** (Yang et al., 2026), an extrapolation-based variant for on-policy distillation.

6.2 Main Results on Reasoning

Table 1 shows that AOPD consistently delivers the best overall reasoning performance across model scales and warm-up settings. For Qwen3-4B-Base with 3k-step SFT warm-up, AOPD already achieves the best average Pass@1, showing that the proposed method is effective in the smaller-scale distillation setting. Moving to Qwen3-8B-Base with the harder 1k-step warm-up initialization, AOPD continues to deliver the best overall performance, whereas standard OPD falls below the warm-up baseline. This result highlights that AOPD is substantially more robust even when the student starts from relatively weak base capabilities. Under the stronger 3k-step warm-up initialization, AOPD further achieves the best overall

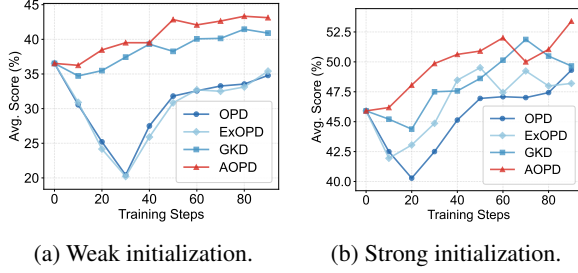


Figure 6: Training dynamics of Qwen3-8B-Base under different initializations.

results, reaching 53.40 average Pass@1 and 67.43 average Pass@4. Extended results are provided in Appendix D.1.

Figure 6 further demonstrates the training dynamics of different methods by plotting the average math score throughout training under different warm-up settings. Across both settings, OPD and ExOPD show a noticeable performance drop at the beginning, which indicates that their optimization is vulnerable to early degradation. This problem is especially severe under the weaker 1K initialization, where the models struggle to recover once performance falls into a poor state, while under the stronger initialization they can recover from degradation but still remain affected by the initial decline. GKD exhibits a more stable trajectory, but its final performance still does not surpass AOPD. In contrast, AOPD improves steadily from the start under both initialization settings and consistently reaches the highest performance ceiling, showing that its asymmetric design not only avoids the early optimization degradation observed in existing baselines but also translates training stability into stronger final reasoning performance. Detailed training dynamics are further shown in Appendix D.2

6.3 Continual Learning and Tool-Use Performance

We evaluate continual learning on ToolAlpaca to test whether the models can acquire new ability while retaining mathematical reasoning performance. Starting from the checkpoints with 3K SFT warm-up in Table 1, we further train each model on ToolAlpaca after math reasoning tasks and report tool-use success rate together with math retention. Table 2 shows that AOPD achieves the best balance between adaptation and retention. It attains a competitive 75.0% success rate while preserving the strongest math performance, improving from 67.43 to 68.04 in average Pass@4. By contrast, OPD,

Table 2: Performance after continual learning.

Method	Tool Use	Math Avg. Pass@4 Before / After	Drop ($\downarrow$)
OPD	65.0	66.64 / 59.15	-7.49
ExOPD	67.0	64.53 / 63.04	-1.49
GKD	76.0	66.96 / 64.54	-2.42
AOPD	75.0	67.43 / 68.04	+0.61

ExOPD, and GKD all suffer math degradation after continual learning, with OPD showing the largest drop. These results suggest that localized teacher guidance enables new ability adaptation without overwriting previously acquired reasoning ability.

6.4 Ablation Studies

6.4.1 Effect of the JSD Parameter β

The parameter β controls the interpolation between student-centric smoothing ($\beta \rightarrow 0$) and teacher-centric guidance ($\beta \rightarrow 1$) during interventions. As discussed in Section 5.1, once intervention is restricted to a teacher-defined support, the correction should remain weighted according to the teacher distribution on that support. Figure 7 shows that

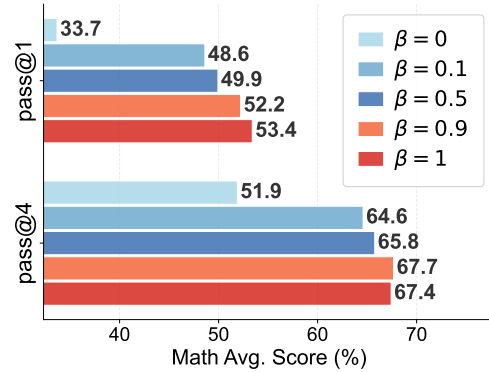


Figure 7: Ablation study on the JSD parameter β .

AOPD attains its highest Pass@1 score at $\beta = 1$, corresponding to strict forward KL. More generally, the results show a clear trend that performance increases as the divergence becomes more biased toward the teacher distribution. This empirical trend further validates the support-consistency principle underlying our design.

6.4.2 Effect of the Top-K

We further study the effect of the teacher support size K in AOPD. As K increases from 8 to 16 and 32 in Figure 8a, the final Pass@1 scores on AIME 2024, AIME 2025, and HMMT 2025(Feb) improve, suggesting that a larger top- K preserves a richer

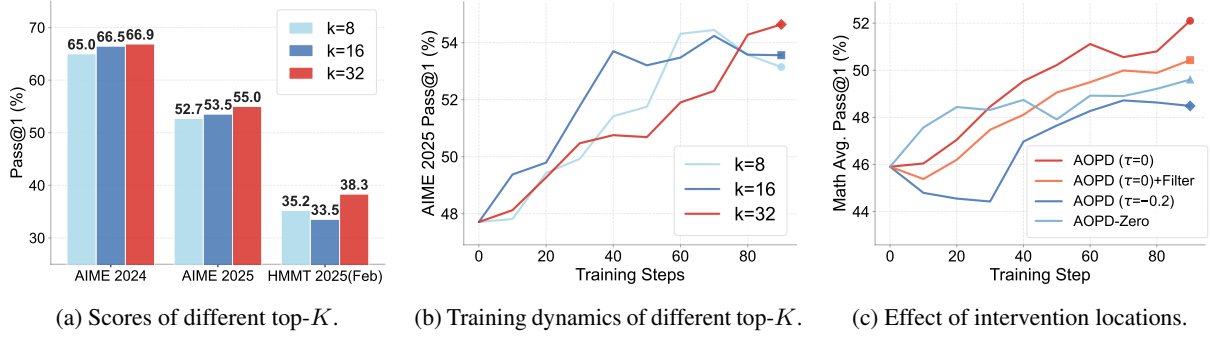


Figure 8: Ablation study on top- K and intervention location.

portion of the teacher distribution and thus provides more complete signals. At the same time, we observe that smaller K often yields larger gains at the early stage of training in Figure 8b, indicating that restricting supervision to a few high-probability teacher tokens produces denser and more concentrated learning signals, which is more beneficial for rapid initial improvement. In contrast, although larger K brings less pronounced early gains, it retains a broader candidate distribution and therefore provides more informative signals in later training, leading to a higher final performance ceiling.

6.4.3 Effect of Intervention Location

The effectiveness of AOPD depends not only on whether KL guidance is used, but also on where the intervention is triggered. We consider three strategies: a token-level threshold τ , which selects low confidence positions for intervention; an accuracy filter, which applies KL guidance only to negative advantage regions from trajectories with incorrect final answers; and AOPD-Zero, which focuses exclusively on neutral regions with $A_t \approx 0$.

Figure 8c shows that standard AOPD with $\tau = 0$ achieves the strongest overall performance, whereas the more conservative setting $\tau = -0.2$ improves more slowly, indicating that shrinking the intervention range weakens the benefit of teacher guidance. Extended studies on τ are presented in Appendix D.3. The accuracy-filtered variant AOPD-Zero remains competitive but consistently trails standard AOPD under the same threshold, suggesting that informative correction signals also arise in correct trajectories. Meanwhile, AOPD-Zero outperforms $\tau = -0.2$, which confirms that neutral regions are a major source of optimization stagnation in OPD. However, focusing only on zero-advantage positions still captures only part of the locations that benefit from intervention. Over-

all, the best results are achieved only by standard AOPD, and any reduction in the set of intervened positions leads to a clear drop in performance.

7 Conclusion

While on-policy distillation is an effective paradigm for transferring reasoning capabilities, its scalar advantage signal does not fully capture the teacher’s richer distributional preferences in difficult regions. In this work, we identify this limitation in non-positive advantage regions and propose Asymmetric On-Policy Distillation, which retains the standard policy gradient update where it is reliable and applies direct teacher distribution alignment where stronger correction is required. Experiments across multiple scales and benchmarks show that AOPD consistently improves mathematical reasoning and remains stable under different model scales and initializations, where standard OPD often struggles. More broadly, our findings suggest that a learning paradigm combining exploitation and imitation can improve both training efficiency and the capabilities of the student model. We discuss limitations in Appendix C and leave extension to broader tasks for future work.

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A Related Work about Joint SFT-RL Optimization

Because on-policy distillation follows a reinforcement learning paradigm, it remains vulnerable to exploration bottlenecks. To alleviate this issue, recent work incorporates supervised learning or expert demonstrations into RL. Some methods dynamically interleave SFT and RL, such as ReLIFT (Ma et al., 2026), SASR (Chen et al., 2025a), and BRIDGE (Chen et al., 2025b), while others unify them in a single-stage objective, including SRFT (Fu et al., 2026) and Prefix-RFT (Huang et al., 2025). A related line instead interprets SFT as off-policy guidance within on-policy RL, as in LUFFY (Yan et al., 2025) and CHORD (Zhang et al., 2026a). In this spirit, AOPD introduces localized soft-label supervised learning to improve exploration efficiency in OPD.

B Extended Analysis on Truncated Teacher Guidance

B.1 Gradient Analysis

In this section, we compare the logit-level gradients induced by forward KL and reverse KL under the same teacher-defined support S_t . For simplicity, $P_S(\cdot | c_t)$ and $P_T(\cdot | c_t)$ denote the normalized student and teacher distributions on S_t . We first consider the forward KL objective

$$\mathcal{L}_t^{\text{FKL}} = D_{\text{KL}}(P_T(\cdot | c_t) \| P_S(\cdot | c_t)) = \sum_{v \in S_t} P_T(v | c_t) \log \frac{P_T(v | c_t)}{P_S(v | c_t)}. \quad (13)$$

Taking derivatives with respect to the student logits, the term $\sum_{v \in S_t} P_T(v | c_t) \log P_T(v | c_t)$ can be treated as a constant. We therefore have

$$\frac{\partial \mathcal{L}_t^{\text{FKL}}}{\partial z_j} = \frac{\partial}{\partial z_j} \left(- \sum_{v \in S_t} P_T(v | c_t) \log P_S(v | c_t) \right) = P_S(j | c_t) - P_T(j | c_t), \quad j \in S_t. \quad (14)$$

We next consider the reverse KL objective

$$\mathcal{L}_t^{\text{RKL}} = D_{\text{KL}}(P_S(\cdot | c_t) \| P_T(\cdot | c_t)) = \sum_{v \in S_t} P_S(v | c_t) \log \frac{P_S(v | c_t)}{P_T(v | c_t)}. \quad (15)$$

Differentiating with respect to z_j gives

$$\frac{\partial \mathcal{L}_t^{\text{RKL}}}{\partial z_j} = P_S(j | c_t) \left(\log \frac{P_S(j | c_t)}{P_T(j | c_t)} - D_{\text{KL}}(P_S(\cdot | c_t) \| P_T(\cdot | c_t)) \right), \quad j \in S_t. \quad (16)$$

The difference between Eq. 14 and Eq. 16 is important in the intervention regime. Forward KL yields a direct distribution-gap correction on the selected support. By contrast, the reverse KL gradient depends on the current student distribution in a more involved way. Its update is explicitly scaled by $P_S(j | c_t)$, and the inner term is also determined by a student-dependent log-ratio relative to the global reverse KL value. As a result, the recovery signal is less direct when a teacher-preferred token has already been strongly suppressed by the student. This comparison explains why forward KL is better suited to the teacher-defined support used in AOPD.

B.2 Training Dynamics

We further examined the training dynamics under different β values and observed an anomalous pattern when the objective tilts toward reverse KL. As shown in Section 6.4, the entropy under reverse KL ($\beta = 0.0$) undergoes a sharp initial surge, followed by an abrupt collapse to near zero. This trajectory indicates that the student rapidly transits into a converged, low-entropy policy. Concurrently, Figure 9b reveals that the proportion of tokens receiving JS divergence guidance plummets at the same entropy collapse phase, implying that most generated positions register positive advantages and exit the intervention regime.

The ablation results in Section 6.4 demonstrate that this apparent convergence coincides with severe degradation in reasoning capability, with $\beta = 0.0$ attaining merely 33.7% Pass@1 compared to 53.4%

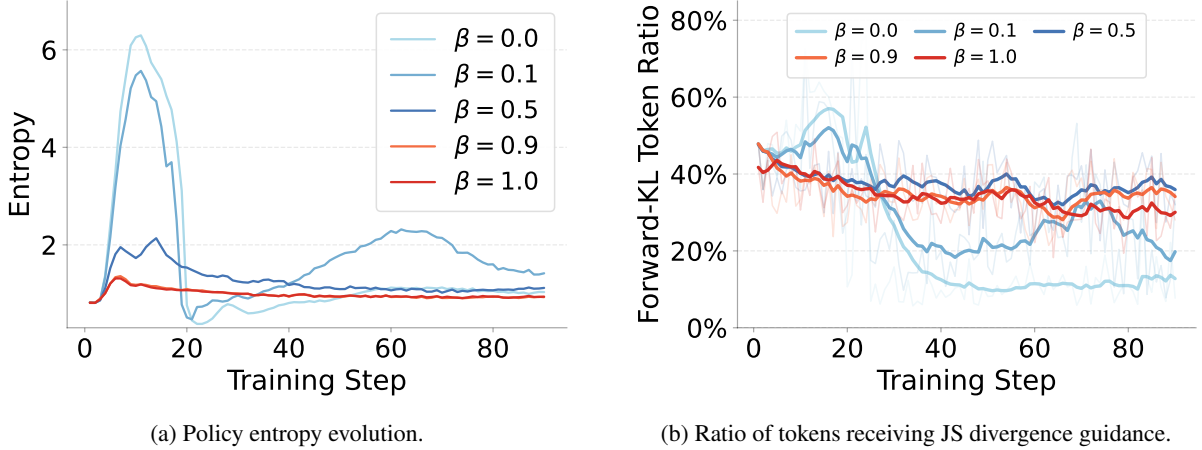


Figure 9: Training dynamics under different β values.

under forward KL. We attribute this to a reward hacking phenomenon in on-policy distillation. The student discovers a mode-collapsed strategy that artificially suppresses the reverse KL value while failing to acquire the teacher’s reasoning patterns. In contrast, forward KL maintains stable entropy and sustained intervention throughout training, yielding both superior and more robust performance.

C Limitations

Our study focuses on teacher-student distillation and evaluates AOPD mainly on mathematical reasoning and sequential tool-use adaptation. Extending the same asymmetric intervention mechanism to self-distillation is a promising direction, but we leave a careful investigation of that setting to future work. More broadly, assessing AOPD across a wider range of tasks and training regimes would further clarify its generality.

D Extended Experiments

D.1 Extended Main Results

We further supplement Table 3 with the results of Qwen3-4B-Base under a 1K-step SFT warm-up initialization. We observe that AOPD still attains the best overall performance, which is consistent with the trend observed in the main text.

Table 3: Supplementary distillation results for Qwen3-4B-Base across multiple math reasoning benchmarks. We compare AOPD against SeqKD, OPD, GKD and ExOPD. **Bold** indicates the best result within each configuration block, while underlined values denote the second best.

Method	AIME 2024		AIME 2025		HMMT 2025(Feb)		Average	
	Pass@1	Pass@4	Pass@1	Pass@4	Pass@1	Pass@4	Pass@1	Pass@4
Qwen3-4B-Base								
– 1K SFT Warm-up								
SFT Checkpoint	36.25	58.61	32.71	47.25	19.17	28.80	29.38	44.89
+ SeqKD	42.50	65.23	40.00	60.09	24.58	38.01	35.69	54.44
+ OPD	46.25	65.99	42.08	60.75	22.50	35.67	36.94	54.14
+ ExOPD	44.58	64.41	41.46	61.70	20.62	32.32	35.55	52.81
+ GKD	<u>50.21</u>	<u>70.47</u>	44.17	<u>63.03</u>	<u>27.50</u>	<u>42.21</u>	<u>40.63</u>	<u>58.57</u>
+ AOPD	51.25	72.01	<u>43.96</u>	63.49	28.33	46.02	41.18	60.51

D.2 Detailed Training Curves

We report the full test-set performance dynamics throughout training for both warm-up settings on Qwen3-8B-Base. All metrics are evaluated every 10 steps on the held-out test sets of AIME 2024, AIME 2025, and HMMT 2025(Feb). Figures 10 traces Pass@1/4/8 across the training trajectory.

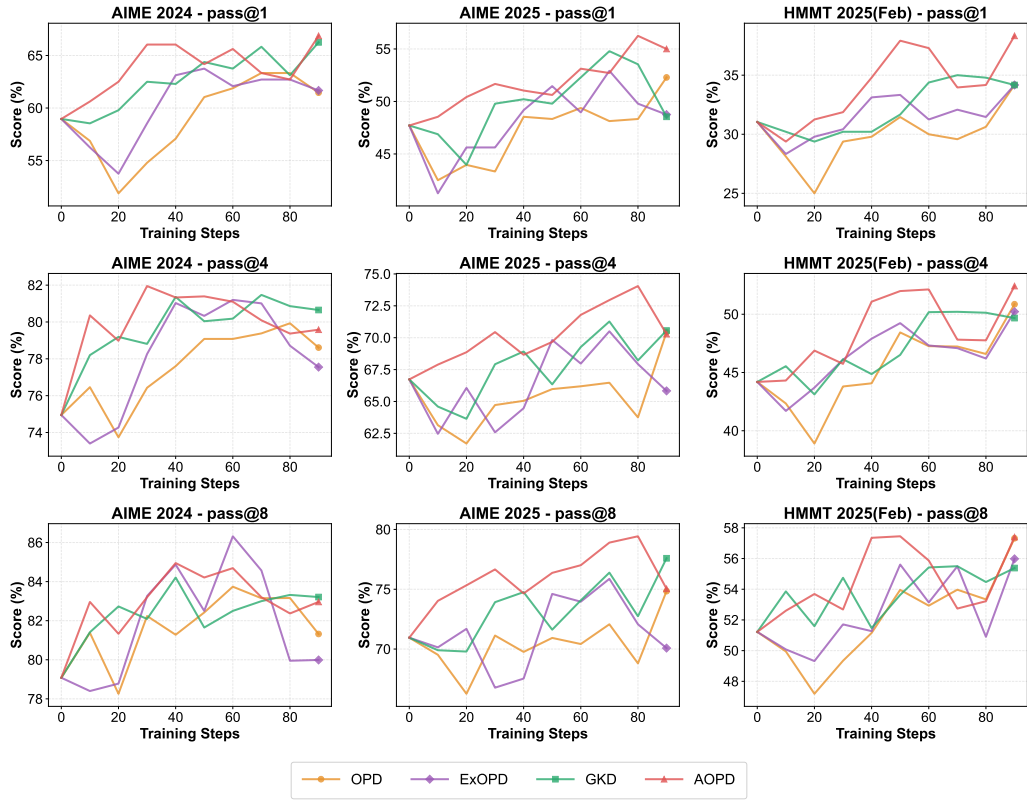
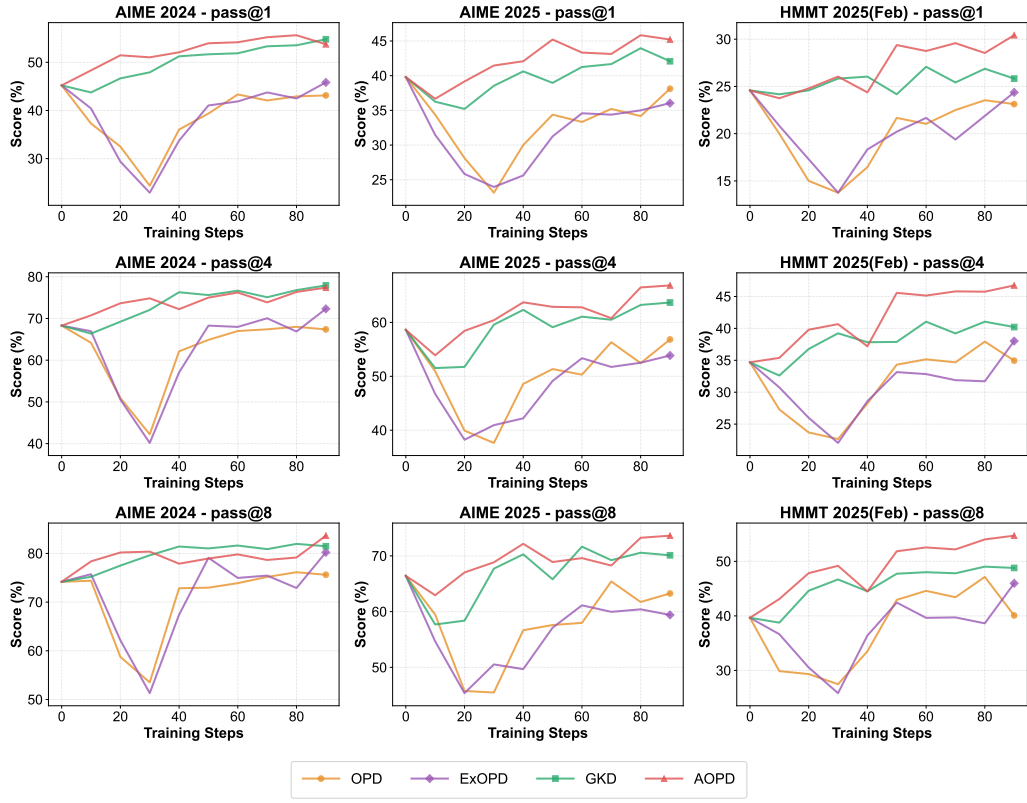


Figure 10: Detailed training dynamics of Qwen3-8B-BASE under weak and strong initialization.

Across both initialization settings, AOPD consistently achieves the highest scores on Pass@1/4/8 across all benchmarks. Under weak initialization, this advantage is particularly pronounced. AOPD maintains steady improvement from the outset, whereas OPD and ExOPD exhibit severe early degradation followed by slow recovery. The substantial performance gap indicates that AOPD remains effective even for students with weaker foundational capabilities. Under strong initialization, the margin between AOPD and baselines narrows, though AOPD still reaches the highest performance ceiling. This convergence suggests that methods such as OPD are more dependent on the student’s initial capability. A stronger base model produces trajectories where student tokens more frequently receive positive advantages from the teacher, thereby reducing the frequency of exploration black holes and alleviating the structural weaknesses of standard negative reinforcement.

D.3 Extended Ablation Studies on Parameter τ

We further conduct an extended study on the intervention threshold τ . As shown in Figure 11a, the mathematical reasoning performance gradually declines as τ decreases, and begins to plateau once τ reaches -0.2 . This trend suggests that reducing τ lowers the proportion of positions that receive forward KL supervision, thereby weakening the teacher-guided correction effect in AOPD. As the intervention becomes increasingly sparse, the model correspondingly moves closer to standard OPD, and the performance eventually stabilizes.

This behavior is further explained by the intervention ratio in Figure 11b. When $\tau \leq -0.2$, the fraction of tokens that trigger the supervision loss drops to below 10% of all generated tokens. As a result, only a very small subset of positions receives explicit teacher guidance, which explains why the performance gap becomes relatively small once τ is reduced beyond -0.2 . These results suggest that the full benefit of AOPD is realized only when teacher guidance spans the entire zero-advantage and negative-advantage regions, whereas reducing forward KL supervision in the negative-advantage region correspondingly weakens the effect of AOPD.

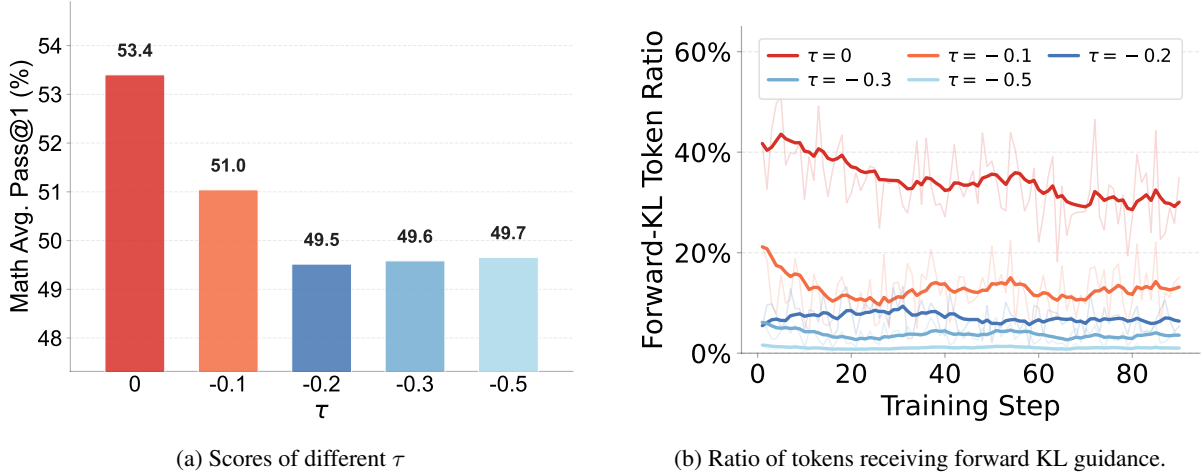


Figure 11: Training dynamics under different τ values.